

Dimension Convergence Model: A Resolution-Dependent Framework for Perceived Dimensionality under Finite Observation

Abstract

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Classical geometry treats dimension as an intrinsic property of objects. However, empirical observation suggests that perceived dimensionality depends on observational resolution: distant objects appear point-like, thin structures appear one-dimensional, and planar surfaces appear two-dimensional despite existing in three-dimensional space.

This work proposes the Dimension Convergence Model, in which dimensionality is treated as an observer-dependent, resolution-constrained quantity. A distinction is introduced between ontological dimension and perceived dimension, and a formal framework is constructed based on finite-resolution measurement and observable information.

Unlike classical fractal dimension, which is typically defined in asymptotic scaling regimes, this model focuses on finite-resolution observational conditions, incorporating anisotropic object structure and detectability constraints. The framework provides a structured interpretation of dimensional reduction and highlights its relevance in both physical observation and complex system modeling.

1 Introduction

Dimension is traditionally defined as the minimum number of coordinates required to specify a point in space:

- 0D — point
- 1D — line
- 2D — surface
- 3D — volume

This definition assumes that dimension is an intrinsic property of objects.

However, empirical observations suggest that perceived dimensionality depends on observational resolution:

- distant stars appear as point sources

- thin wires appear one-dimensional
- surfaces appear two-dimensional

This work proposes a framework in which dimension is treated as an observable quantity under finite resolution.

2 Ontological and Perceived Dimension

We distinguish two notions:

2.1 Ontological Dimension

The intrinsic dimensionality of an object in physical space:

$$D_{ont} = 3$$

2.2 Perceived Dimension

The effective dimensionality observable under finite resolution:

- distant star $\rightarrow D_p \approx 0$
- thin wire $\rightarrow D_p \approx 1$
- planar surface $\rightarrow D_p \approx 2$

$$0 \leq D_p \leq D_{ont}$$

3 Measurement Framework

3.1 Multi-Channel Resolution

An observer is modeled as having multiple measurement channels:

$$\{R_1, R_2, \dots, R_n\}$$

Each channel has resolution R_i and weight w_i .

We define effective resolution:

$$R_{\text{eff}} = \left(\sum_{i=1}^n w_i R_i^{-p} \right)^{-1/p}$$

This formulation corresponds to a standard L_p -norm aggregation and is adopted here as a convenient representation of multi-channel resolution rather than a novel construction.

3.2 Anisotropic Object Structure

Objects are characterized by directional scales:

$$\mathbf{S} = (S_1, S_2, \dots, S_{D_{ont}})$$

This allows representation of anisotropic structures such as wires and surfaces.

3.3 Observable Structure Count

We define:

$$N_{\text{geom}} = \prod_{i=1}^{D_{ont}} \max\left(1, \frac{S_i}{R_{\text{eff}}}\right)$$

Here N_{geom} is interpreted as a continuous measure of observable structural capacity rather than a discrete count, allowing smooth transitions under finite resolution.

3.4 Detectability Constraint

We introduce a detectability factor:

$$f(R_{\text{eff}}) = \frac{1}{1 + (R_{\text{eff}}/R_c)^\alpha}$$

The effective observable structure becomes:

$$N_{\text{eff}} = N_{\text{geom}} \cdot f(R_{\text{eff}})$$

4 Resolution-Dependent Dimension

We define perceived dimension as:

$$D_p = \begin{cases} 0, & R_{\text{eff}} \geq S_{\text{max}} \text{ or } N_{\text{eff}} < 1 \\ \frac{\log N_{\text{eff}}}{\log(S_{\text{max}}/R_{\text{eff}})}, & \text{otherwise} \end{cases}$$

where $S_{\text{max}} = \max(S_i)$.

5 Limiting Behavior

- As $R_{\text{eff}} \rightarrow 0$: $D_p \rightarrow D_{ont}$
- As $R_{\text{eff}} \rightarrow \infty$: $D_p \rightarrow 0$

6 Examples

6.1 Thin Wire

If $S_1 \gg S_2, S_3$ and $R_{\text{eff}} > S_2, S_3$:

$$D_p \approx 1$$

6.2 Planar Surface

If $S_1, S_2 \gg S_3$ and $R_{\text{eff}} > S_3$:

$$D_p \approx 2$$

6.3 Sphere

If $S_1 = S_2 = S_3$, the object remains effectively three-dimensional until resolution approaches object scale, after which:

$$D_p \rightarrow 0$$

7 Relation to Existing Frameworks

This work relates to fractal dimension, coarse-graining, and information theory. It does not replace these frameworks but provides a finite-resolution, anisotropic extension emphasizing observer-dependent dimensional perception.

8 Implications

8.1 Physical Observation

Dimensionality depends on measurement resolution.

8.2 Measurement Systems

Different observers may assign different dimensions.

8.3 Complex Systems

Dimensional mismatch may arise when low-dimensional models are applied to high-dimensional systems.

9 Limitations

- Simplified detectability model
- Parameters R_c and α require empirical grounding
- Framework is interpretive and operational rather than fundamental

10 Conclusion

The Dimension Convergence Model interprets dimensionality as a resolution-dependent observable. By incorporating anisotropic structure and detectability constraints, the model provides a structured framework for understanding dimensional perception under finite observation.